

C2-05 Module QA Report

Python Automation for Analysts - beginner-first RC1 - 2026-09-10

VERDICT

PASS for available module-level static, PDF-render, notebook-structure/syntax, archive CRC, Excel-template and backward-compatible runtime-smoke QA. Target Python 3.14.7 / pandas 3.0.5 live execution and the unified Course 2A Mentor remain pending. Course 2A is NOT FINAL.

LEARNER-FACING REBUILD

11 canonical PYA lessons rebuilt into a 51-page teacher-led book using the full M01 grammar. Exact starts reconcile 11/11: PYA1 p6, PYA2 p10, PYA3 p14, PYA4 p18, PYA5 p22, PYA6 p26, PYA7 p30, PYA8 p34, PYA9 p38, PYA10 p42, PYA11 p46.

PRACTICE / ASSESSMENT

12-file practice pack, full P-PYA1..P-PYA11 notebook, Weekly Reporting Mini-Lab, 12-page protected review, five targeted remediation cards, and three fresh automation Gates: Retail, Support and Subscription Billing.

NO HAPPY-PATH ONLY

Practice injects backup/schema drift, duplicate key, unmatched key, malformed numeric, nested/optional JSON and offline API data. Critical checks must withhold publication.

C2-05 QA - Runtime, notebook and file integrity

Defects were caught before byte freeze

RUNTIME SMOKE

Protected smoke PASS under Python 3.13.5 / pandas 2.2.3 with FutureWarning promoted to error. It validates approved-file discovery, typed calculations, duplicate-key detection, nested JSON count, offline API fixture, Excel round-trip and run-manifest output. This is backward-compatible smoke evidence only.

DEFECTS CAUGHT + FIXED

QA first caught a one-cent floating-point comparison caused by binary representation; the check now rounds to cents before applying ≤ 0.01 tolerance. The next run caught a pandas FutureWarning in an external-grouper pattern; the QA reference now creates an explicit grouping column. Final warning-as-error smoke PASS.

NOTEBOOK / ARCHIVE QA

All five notebooks parse as nbformat and every code cell compiles as Python syntax. Practice ZIP = 12 entries CRC PASS; Gate A/B/C = 4 entries each CRC PASS. The Excel reference template reopens through artifact_tool with zero formula-error matches and passed visual render inspection.

C2-05 QA - PDF, sources and release boundary

What is frozen versus what remains a Course 2A-level gate

PDF QA

Eight learner/protected PDFs totaling 79 pages were rendered at 200 DPI. All eight preflight as openable, unencrypted and text-based. Three contact sheets were visually inspected with no clipping, overlap, broken glyph or page-route drift observed.

SOURCE / VIDEO QA

17 source/video registry rows across 17 unique URLs. The five baseline registered resources are preserved. Current official Python/pandas/Requests documentation controls behavior; videos are supplemental and external required dependency = 0.

TARGET-VERSION BOUNDARY

Current curriculum references target Python 3.14.7 and pandas 3.0.5. Because this build runtime is older, target-version execution remains PENDING rather than being inferred from documentation or backward smoke success.

COURSE / PARALLEL BOUNDARY

C2-05 is optional C2A depth and is not a prerequisite for C2B/C3. The unified Mentor and whole-course route simulation remain course-level release gates. Shared GitHub/backup integration is untouched.